

Exploring Multilingual Sociocultural Bias in Generating Children’s Stories

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Abstract

Large language models (LLMs) are increasingly used for creative and educational applications, including personalized children’s storytelling. Prior work such as Biased Tales has shown that sociocultural variables embedded in prompts systematically influence narrative content, often reinforcing stereotypical patterns. However, the study focuses primarily on English narratives and does not examine how bias patterns vary across languages or how alignment training interacts with multilingual generation.

In this work, we extend the Biased Tales framework along two dimensions: (1) multilingual narrative generation and (2) comparison between base pretrained and RLHF-aligned models. We translate demographic prompt templates into additional languages using a controlled glossary and back-translation verification protocol to ensure preservation of meaning and demographic labels. We then analyze character-centric traits, contextual settings, demographic predictability, and narrative diversity across languages and training stages. Our study aims to investigate how language and alignment jointly influence sociocultural bias in children’s storytelling models.

1 Introduction

Large language models (LLMs) are increasingly deployed in creative and educational contexts, including personalized children’s storytelling. Narrative content plays a significant role in shaping perceptions of gender, culture, and identity. The Biased Tales study demonstrated that sociocultural variables such as gender, nationality, and religion systematically influence generated stories in English.

However, two important questions remain underexplored. First, how do these bias patterns manifest across different languages? Second, how does

alignment training influence sociocultural bias in multilingual narrative generation?

Modern LLMs are trained in multiple stages, including large-scale pretraining followed by alignment using reinforcement learning from human feedback (RLHF). While alignment aims to optimize helpfulness and safety, its impact on cross-linguistic sociocultural representation remains unclear.

In this work, we aim to explore the aspects like:

How do language and alignment jointly influence sociocultural bias in children’s story generation?

We extend the Biased Tales framework by incorporating multilingual prompt evaluation and comparing base and RLHF-aligned models. Our goal is to explore whether sociocultural bias patterns remain consistent across languages, vary across linguistic contexts, or interact with alignment training in different ways.

2 Related Work

2.1 Sociocultural Bias in Narrative Generation

The Biased Tales dataset introduced a structured framework for measuring character-centric and context-centric bias in children’s stories. Their analysis demonstrated systematic gendered and cultural patterns in English narratives. However, the study primarily focused on English and did not examine multilingual variation.

2.2 Alignment and RLHF

Alignment methods such as reinforcement learning from human feedback (RLHF) are widely used to improve model safety and adherence to social norms. While alignment has been studied in toxicity and instruction-following contexts, its impact on multilingual sociocultural representation in creative generation remains underexplored.

2.3 Multilingual Bias in LLMs

Recent work suggests that bias patterns may differ across languages due to variations in training data distributions and cultural contexts. However, systematic multilingual evaluation of narrative bias remains limited. Our work addresses this gap.

3 Experimental Design

3.1 Model Comparison

We compare two training stages within the same model family:

- **Base Model:** Pretrained using next-token prediction without alignment.
- **RLHF-Aligned Model:** Further optimized using reinforcement learning from human feedback.

This comparison allows us to analyze how alignment interacts with multilingual generation.

3.2 Multilingual Prompt Framework

We adopt controlled demographic prompt templates inspired by Biased Tales. Prompts vary across selected sociocultural axes:

- Child gender (daughter, son)
- Parent nationality group
- Parent religion

We translate prompt templates into one or two additional languages. As of now, we aim to include Hindi and Telugu as additional languages. To ensure translation accuracy (or consistency), we use:

- A fixed glossary for demographic labels (e.g., father, daughter, religion terms).
- Round-trip back-translation to verify meaning preservation, where translated prompts are independently translated back into English and compared with the original prompts to ensure that demographic labels, role information, and instructional intent remain unchanged.
- Human spot-checking using a structured validation checklist.

Example English prompt:

“I am a Muslim father. Write a bedtime story for my daughter.”

3.3 Dataset Generation

For each language and demographic configuration, we generate multiple stories per model using identical decoding parameters. This results in parallel datasets across:

- Language
- Training stage (base vs RLHF)
- Demographic condition

4 Bias Measurement Framework

4.1 Character-Centric Analysis

We extract protagonist attributes from each story and categorize them into physical, emotional, mental, and moral traits. Trait distributions are compared across languages and training stages.

4.2 Context-Centric Analysis

We analyze narrative settings including geographic environments, urban context, and socioeconomic indicators. Cross-linguistic differences in contextual representation are examined.

4.3 Predictability Analysis

Following the Biased Tales framework, we train classifiers to predict demographic attributes (e.g., gender, religion, nationality) from story text representations. We conduct this analysis separately for each language and training stage.

Higher classification accuracy indicates stronger embedding of demographic signals in narratives. Comparing predictability across languages allows us to measure cross-linguistic variation in demographic signal strength, while comparison across training stages reveals how alignment influences such embedding.

4.4 Multilingual Bias Evaluation

We apply the same bias measurement framework independently within each language. Our evaluation consists of three complementary components:

- **Trait-Level Analysis:** We use an LLM-based extraction procedure to identify and categorize protagonist attributes (physical, emotional, mental, moral) from each story. Trait distributions are then compared across demographic conditions.

- **Context-Level Analysis:** Again, we use LLM-assisted extraction to identify narrative settings and environmental descriptions (e.g., geographic or cultural context) and analyze their distribution across demographic groups.
- **Predictability Analysis:** We train language-specific classifiers to predict demographic attributes from story text, measuring the strength of demographic signal embedding.

By combining LLM-based descriptive extraction with demographic predictability measures, we obtain both qualitative and quantitative signals of sociocultural bias across languages and training stages.

5 Expected Outcomes and Analysis

We hypothesize that multilingual and alignment effects may interact in multiple ways:

- Bias intensity may vary across languages due to cultural context.
- RLHF alignment may reduce explicit stereotypes in high-resource languages.
- Alignment effects may be weaker or inconsistent in non-English languages.
- Cultural settings may shift differently across languages.
- Demographic predictability may vary across language-model combinations.

By jointly analyzing language and training stage, we aim to characterize how sociocultural bias propagates and transforms across linguistic contexts.

6 Discussion

This work provides insight into how language and alignment jointly shape sociocultural representation in narrative generation. If bias patterns differ significantly across languages, this suggests that multilingual data distributions play a critical role in bias propagation. If alignment effects vary by language, this may indicate uneven safety generalization across linguistic contexts.

Our findings will contribute to understanding cross-linguistic bias dynamics in aligned language models.

7 References

References

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